

Project Overview

Vision Sports Academy is a dedicated athlete development center located in Akkaraipattu, Ampara, currently training over 100 students across various age groups (Under-08 to Under-18) with a staff of 8 coaches. Founded by Mr. A. Musaraf Mohamed, the academy's primary mission is to provide quality football training, promote discipline, and create opportunities for players to progress to higher competitive levels.

Currently, the academy relies heavily on fragmented, manual workflows. Data collection and registration are handled via Google Forms and paper, schedules and announcements are communicated exclusively through WhatsApp, and attendance and fee payments are recorded manually by management and coaches. This disconnected approach has led to significant administrative bottlenecks, including the difficulty of tracking outstanding monthly payments, a high risk of lost or duplicated records, and an inability to maintain a centralized history of player progress and attendance.

To resolve these inefficiencies, this project aims to design and develop a centralized, web-based **Sports Academy Management System**. This solution will digitize and automate the academy's core operations. It will feature dedicated modules for User Profile Management, Training Schedule & Attendance tracking, and Billing & Reporting. By transitioning to this system, Vision Sports Academy will reduce manual paperwork, improve communication between administrators, coaches, and parents, and establish a highly organized digital foundation to support its future growth.

Functional Requirements

Functional requirements (FRs) describe the specific services the system must provide and how it will react to its inputs. Based on the stakeholder goals identified during the elicitation process, the following system requirements have been defined and partitioned into three main core modules, alongside one optional evaluation module.

FR0.1: The system shall provide a public-facing home page displaying the academy's vision, motto, and promotional group photographs to visitors.

1. User & Profile Management Module (Member 1 Scope)

This feature set allows the academy to manage all human actors within the system, replacing manual registration forms.

- **FR1.1:** The system shall allow the Administrator to create, read, update, and delete (CRUD) Player profiles, including capturing unique IDs, medical information, and kit sizes.
- **FR1.2:** The system shall allow the Administrator to register Parent/Guardian profiles and link a single parent to multiple player profiles.
- **FR1.3:** The system shall allow the Administrator to register Coach profiles, recording their coaching licenses and assigned groups.
- **FR1.4:** The system shall require all users to authenticate using a valid username and password before granting access to system features.
- **FR1.5:** The system shall allow the Administrator to update the status of any account (e.g., Active, Inactive, Suspended).
- **FR1.6:** The system shall allow administrators and coaches to search for specific players by name, unique ID, or age group.

2. Training Schedule & Attendance Module (Member 2 Scope)

This feature set digitizes the academy's calendar and daily tracking, replacing WhatsApp scheduling and manual attendance logs.

- **FR2.1:** The system shall allow the Administrator to schedule new training sessions by assigning a date, start/end time, ground location, coach, and target player group.
- **FR2.2:** The system shall display a centralized schedule calendar visible to Administrators, Coaches, Players, and Parents.
- **FR2.3:** The system shall allow Coaches to mark attendance for their assigned training sessions using predefined statuses (Present, Absent, Late, Excused).
- **FR2.4:** The system shall allow the Administrator to edit submitted attendance records if corrections are required.
- **FR2.5:** The system shall calculate and display the overall attendance percentage for each individual player.

3. Billing, Payments & Reporting Module (Member 3 Scope)

This feature set automates the academy's financial tracking and administrative reporting.

- **FR3.1:** The system shall allow the Administrator to generate monthly training fee records and one-time registration fees for players.
- **FR3.2:** The system shall allow the Administrator to update and track the payment status of each player (Paid, Pending, Overdue).
- **FR3.3:** The system shall allow Parents and Players to view their payment status and historical payment records.
- **FR3.4:** The system shall automatically trigger payment reminder notifications to Parents/Players for pending or overdue fees.
- **FR3.5:** The system shall allow the Administrator to generate and download system reports, including player lists, attendance summaries, and outstanding payment reports.

4. Player Performance Evaluation (Optional Feature)(No Member)

As requested, this functionality is included, but due to insufficient members, we listed it as an optional feature

- **FR4.1:** The system should allow Coaches to input performance scores (e.g., passing, shooting, fitness) for players within their assigned groups.
- **FR4.2:** The system should allow Coaches to attach qualitative text comments and development goals to a player's performance record.
- **FR4.3:** The system should allow Parents and Players to view their accumulated performance progress over time.

Non-Functional Requirements

Non-functional requirements (NFRs) characterize the system's overall behavior, qualities, and constraints rather than specific user features. Based on the stakeholder elicitation and the specified classification criteria, the following NFRs have been established.

1. Usability

- **NFR1.1 (Mobile-Friendliness):** The system shall be highly responsive and mobile-friendly, ensuring seamless access across both desktop computers and mobile phones.
- **NFR1.2 (Browser Compatibility):** The system's user interface shall be optimized for performance on the Google Chrome web browser.
- **NFR1.3 (Aesthetics):** The user interface, particularly the public home page, should be visually appealing and clearly reflect the academy's branding and professionalism.

2. Security

- **NFR2.1 (Authentication):** The system shall require all users to authenticate using a registered username (or email) and password.
- **NFR2.2 (Access Control):** The system shall prevent unauthorized users from viewing information and strictly restrict sensitive academy financial/user data to administrator accounts.
- **NFR2.3 (Auditability):** The system shall maintain an audit log recording all important modifications made by users within the system.

3. Reliability

- **NFR3.1 (Offline Capability):** In the event of network failure, the system shall support temporary offline data storage and automatically synchronize the data with the server once the internet connection is restored.
- **NFR3.2 (Data Backup):** The system database shall execute automated daily backups to prevent data loss.

4. Performance

- **NFR4.1 (Response Time):** The system shall maintain very fast response speeds during standard operations, as system latency is highly critical to the client's daily administrative tasks.

5. Maintainability

- **NFR5.1 (Extensibility):** The system codebase shall be modular and apply Object-Oriented design principles to easily support future requirements, such as adding new sports or integrating additional academy branches.

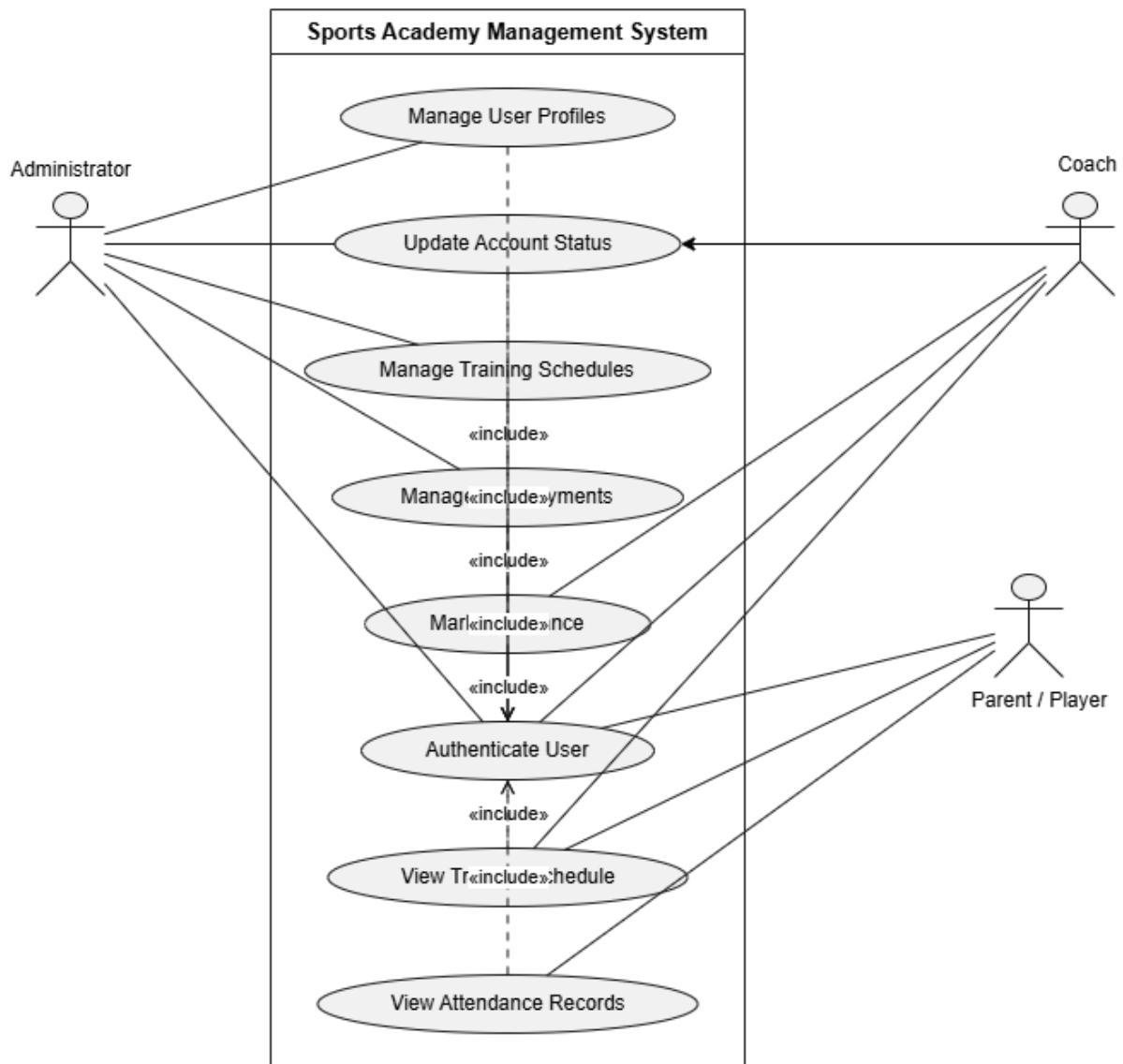
6. Interoperability

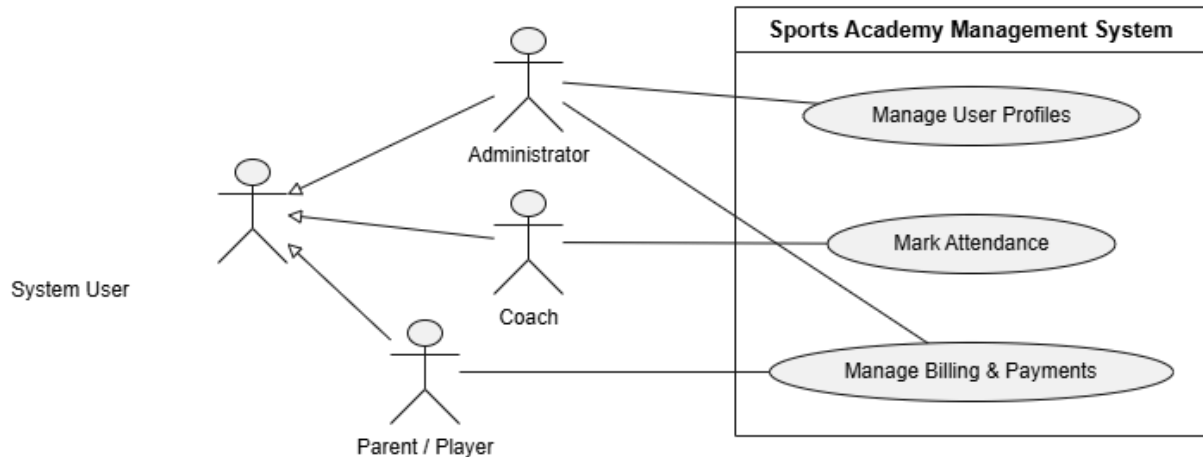
- **NFR6.1 (External Communication):** The system should be capable of interfacing with standard communication APIs (e.g., WhatsApp/SMS gateways) to deliver automated schedule and payment notifications.

7. Constraints (Design & Implementation)

- **NFR7.1 (Framework Architecture):** The solution shall be implemented as a Spring Web Application utilizing Java Spring Boot.
- **NFR7.2 (Backend Structure):** The backend shall be developed using Spring MVC and expose a RESTful API for communication with the frontend UI.

UML Use-Case Diagram





Use-Case Scenarios

Use Case 1: Manage User Profiles (Member 1 Scope)

Field	Description
Use Case ID	UC-01
Name	Manage User Profiles
Primary Actor	Administrator
Summary	The administrator creates a new system account (Player, Parent, or Coach) to replace the manual Google Forms registration process.
Preconditions	The Administrator is successfully authenticated and logged into the system.

Field	Description
Main Flow	1. Administrator selects "Manage Users" from the main dashboard. 2. Administrator chooses to "Register New Player". 3. Administrator enters the player's personal details, medical information, and kit sizes. 4. System validates the entered information. 5. System generates a Unique ID and saves the new profile. 6. System displays a success message to the Administrator.
Postconditions	The new user profile is permanently stored in the database and the account status is set to "Active".

Use Case 2: Mark Attendance (Member 2 Scope)

Field	Description
Use Case ID	UC-02
Name	Mark Attendance
Primary Actor	Coach
Summary	A coach records the daily attendance of players for a specific scheduled training session.
Preconditions	The Coach is authenticated, and a training session has been scheduled by the Administrator for the Coach's assigned group.
Main Flow	1. Coach selects "View Training Schedule" and clicks on today's session. 2. System displays the roster of players assigned to that specific group. 3. Coach marks the attendance status for each player (Present, Absent, Late, Excused).

Field	Description
	4. Coach clicks "Submit Attendance". 5. System calculates and updates the overall attendance percentage for each player. 6. System saves the record and displays a confirmation message.
Postconditions	The session's attendance is securely recorded, and the updated attendance percentages are visible to the Administrator and Parents.

Use Case 3: Manage Billing & Payments (Member 3 Scope)

Field	Description
Use Case ID	UC-03
Name	Manage Billing & Payments
Primary Actor	Administrator
Summary	The administrator updates a player's monthly training fee status from "Pending" to "Paid" upon receiving payment.
Preconditions	The Administrator is authenticated, and a monthly fee record has already been generated for the target player.
Main Flow	1. Administrator navigates to the "Manage Fee Payments" module. 2. Administrator searches for a specific player by Name or Unique ID. 3. System displays the player's pending monthly fees.

Field	Description
	4. Administrator selects the pending month and clicks "Mark as Paid". 5. System updates the payment status in the database. 6. System triggers an automated payment confirmation notification to the linked Parent's account.
Postconditions	The financial record is updated to "Paid", and the pending balance is cleared from the Parent's view.